SQL Data Analysis Exercises

20 Beginner Queries for DuckDB & Relational Databases

Using the Instacart sample dataset (Aisles, Departments, Orders, Products, and Order_Products), complete the following analysis tasks using SQL:

Level 1: Basic Data Exploration

1. Retrieve the first 10 rows from the `products` table to understand the available columns.
2. Find all unique department names from the `departments` table.
3. Count the total number of orders placed in the `orders` dataset.
4. List all product names found specifically in Aisle ID 61.
5. Display all orders sorted by the hour of the day they were placed (highest to lowest).

Level 2: Aggregation & Grouping

1. Calculate the total number of products associated with each `department_id`.
2. Find the average number of days that pass between a user's orders (`days_since_prior_order`).
3. Count how many orders were placed on each day of the week (`order_dow`).
4. Identify the single busiest hour of the day for shopping based on the total number of orders.
5. Find the maximum value in the `add_to_cart_order` column to see the largest single-order cart size.

Level 3: Joining Multiple Tables

1. Create a list of product names alongside their respective department names.
2. Retrieve a list showing `order_id` and the names of the products contained in each order.
3. List all products and the name of the aisle they belong to.
4. Calculate the total number of items sold per department name.

5. Display `user_id`, `order_id`, and `product_name` for the first 20 records in the dataset.

Level 4: Intermediate Analysis

1. Calculate the overall reorder rate (the average of the `reordered` column).
2. Find the top 5 most popular products based on the number of times they appear in the `order_products` table.
3. Identify the top 10 users who have placed the highest number of total orders.
4. Find the top 5 products that are most frequently added to the cart first (where `add_to_cart_order = 1`).
5. Categorize orders into 'Weekend' (Days 0 and 1) vs 'Weekday' (Days 2-6) and count the total orders for each category.

Practice Dataset: Instacart Market Basket Analysis Sample